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Inventor(s)

Lee; Bradley J.

UNIVERSAL ROLLER BRACKET FOR A MOVABLE BARRIER

Abstract

A universal roller bracket for a movable barrier having a surface, the universal roller bracket comprising: a first bracket section configured to be coupled to the movable barrier; a first arm extending from the first bracket section; and a plurality of roller tubes supported by the first arm and spaced from the surface of the movable barrier by the first arm, each roller tube of the plurality of roller tubes configured to receive a roller stem, each roller tube being spaced a different distance from the surface of the movable barrier.

Inventors: Lee; Bradley J. (Arlington, TX)

Applicant: OVERHEAD DOOR CORPORATION (LEWISVILLE, TX)

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to the field of movable barriers. In particular, a universal roller bracket spaces rollers from a movable barrier at differing distances.

BACKGROUND

[0002] Movable barriers, such as upward-acting sectional or single panel garage doors, residential and commercial rollup doors, and slidable and swingable gates, are used to alternatively allow and restrict entry to building structures and property. An upward-acting slidable barrier typically includes rollers positioned on either side of the barrier which roll along a track. Each roller is secured to the barrier with a roller bracket. In a closed position, the barrier is positioned vertically and the track is angled from the barrier extending upward. In this way, the distance between the barrier and the track increases with vertical height. The angle of the track with respect to the barrier ensures that the barrier is pulled away from the jamb of the barrier as the barrier is slid upward during opening.

[0003] The roller brackets of the barrier are typically selected to maintain the angle between the track and the barrier in a closed position. As a result, each bracket is individually sized to space the roller from the movable barrier by a fixed, set distance. For example, one bracket at the bottom of the barrier may space a roller from the barrier by a minimal amount, while other brackets above the bottom bracket may space rollers from the barrier by other fixed amounts. The movable barrier typically includes several differently sized roller brackets to achieve this result. Specifically, each roller bracket of the barrier has a roller stem tube mounting location unique to the roller brackets at any other section of the barrier. In the case of a four-section barrier, up to five unique roller brackets may be required, with at least one at each section joint and a bracket at the top and bottom of the barrier to provide the necessary graduated track angle. Multiple unique brackets typically requires that servicers have several different bracket types, increasing inventory tracking complexity as well as complexity and costs of servicing barriers or maintaining on-hand inventory.

SUMMARY

[0004] In some example aspects, the present disclosure is directed to a universal roller bracket for a movable barrier. The universal roller bracket includes a surface and a first bracket section configured to be coupled to the movable barrier; a first arm extending from the first bracket section; and a plurality of roller tubes supported by the first arm and spaced from the surface of the movable barrier by the first arm, each roller tube of the plurality of roller tubes configured to receive a roller stem, each roller tube being spaced a different distance from the surface of the movable barrier.

[0005] In some aspects, the first bracket section is configured to be slidably coupled to the surface of the movable barrier. In some aspects, the first bracket section includes a protrusion configured to be positioned within a complementary groove of the surface of the movable barrier. In some aspects, the universal roller bracket includes a second bracket section spaced from the first bracket section, the second bracket section configured to be fixedly coupled to the surface of the movable barrier with a fastener. In some aspects, the distance of the roller stem from the surface of the movable barrier is based on an installation height of the roller stem when the movable barrier is in a closed position. In some aspects, the different distances of each roller tube correspond to a linear or non-linear relationship. In some aspects, the universal roller bracket includes a second arm, the first arm and the second arm forming a gap between the plurality of roller tubes and the surface of the movable barrier. In some aspects, the first bracket section includes a protrusion configured to be inserted into a complementary groove of the surface of the movable barrier.

[0006] In some example aspects, the present disclosure is directed to a movable barrier system. The movable barrier system includes a first section of a movable barrier comprising an inner surface and an outer surface, the inner surface comprising a retaining groove; a first universal roller bracket comprising: a first bracket section comprising a protrusion configured to be positioned within the

retaining groove of the first section; and a first plurality of roller tubes spaced from the inner surface, wherein each roller tube is positioned a different distance from the inner surface; and a roller comprising a roller stem, wherein the roller stem is configured to be positioned within one of the plurality of roller tubes.

[0007] In some aspects, the movable barrier is configured to be substantially vertical when in a closed position when the roller stem is positioned within the one roller tube of the plurality of roller tubes. In some aspects, the first universal roller bracket includes a second bracket section configured to be fixedly coupled to the first section. In some aspects, the movable barrier system includes a second section of the movable barrier comprising an inner surface and an outer surface; and a second universal roller bracket including a second plurality of roller tubes, the second universal roller bracket configured to be coupled to the second section, wherein each roller tube of the second plurality of roller tubes is positioned a different distance from the inner surface of the second section, and wherein the distance of each roller tube of the second plurality of roller tubes from the inner surface of the second section is different than the distance of each roller tube of the first plurality of roller tubes of the first universal roller bracket from the inner surface of the first section. In some aspects, the second section is positioned above the first section and the second universal roller bracket is positioned above the first universal roller bracket when the movable barrier is in a closed position.

[0008] In some example aspects, the present disclosure is directed to a method of assembling a movable barrier system. The method includes positioning a roller within a track of the movable barrier system; slidably coupling a first surface of a universal roller bracket to a section of a movable barrier, the universal roller bracket comprising a plurality of roller tubes spaced from the section of the movable barrier, wherein each roller tube of the plurality of roller tubes is spaced from the section by a different distance; sliding the universal roller bracket to a position near an end of the section of the movable barrier; inserting a roller stem of the roller within one roller tube of the plurality of roller tubes; and fixedly coupling a second surface of the universal roller bracket to the section of the movable barrier.

[0009] In some aspects, the method includes fixedly coupling the first surface of the universal roller bracket to the section of the movable barrier. In some aspects, slidably coupling the first surface of the universal roller bracket to the section of the movable barrier includes positioning a protrusion of the universal roller bracket within a corresponding groove of the section of the movable barrier. In some aspects, coupling the second surface of the universal roller bracket to the section of the movable barrier includes coupling the second surface of the universal roller bracket to the section of the movable barrier with a fastener. In some aspects, the one roller tube of the plurality of roller tubes into which the roller stem is inserted is selected based on an installation height of the universal roller bracket on the section of the movable barrier. In some aspects, the installation height of the universal roller bracket is determined when the movable barrier is in a closed position.

[0010] It is to be understood that both the foregoing general description and the following drawings and detailed description are exemplary and explanatory in nature and are intended to provide an understanding of the present disclosure without limiting the scope of the present disclosure. In that regard, additional aspects, features, and advantages of the present disclosure will be apparent to one skilled in the art from the following. One or more features of any implementation or aspect may be combinable with one or more features of other implementation or aspect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings illustrate implementations of the systems, devices, and

methods disclosed herein and together with the description, serve to explain the principles of the present disclosure.

[0012] FIG. 1 is a perspective illustration of a movable barrier system, according to aspects of the present disclosure.

[0013] FIG. 2 is a side view of a movable barrier in a closed position along a track with universal roller brackets, according to aspects of the present disclosure.

[0014] FIG. 3 is a perspective view of a section of a movable barrier, a universal roller holder, and a roller, according to aspects of the present disclosure.

[0015] FIG. 4 is a perspective view of a section of a movable barrier, a universal roller holder, and a roller, according to aspects of the present disclosure.

[0016] These Figures will be better understood by reference to the following Detailed Description.
DETAILED DESCRIPTION

[0017] For the purpose of promoting an understanding of the principles of the present disclosure, reference will now be made to the implementations illustrated in the drawings and specific language will be used to describe them. It will nevertheless be understood that no limitation of the scope of the disclosure is intended. Any alterations and further modifications to the described devices, instruments, methods, and any further application of the principles of the present disclosure are fully contemplated as would normally occur to one skilled in the art to which the disclosure relates. In addition, this disclosure describes some elements or features in detail with respect to one or more implementations or Figures, when those same elements or features appear in subsequent Figures, without such a high level of detail. It is fully contemplated that the features, components, and/or steps described with respect to one or more implementations or Figures may be combined with the features, components, and/or steps described with respect to other implementations or Figures of the present disclosure. For simplicity, in some instances the same or similar reference numbers are used throughout the drawings to refer to the same or like parts.

[0018] In some aspects, the present disclosure relates to a universal roller bracket used to hold rollers of an upward-acting movable barrier. The universal roller bracket includes multiple roller tubes which may receive a stem of a roller. Each roller tube is arranged to be a different distance from a plane formed by the movable barrier. Accordingly, a user can choose which roller tube to use for a particular roller based on the location of the universal roller bracket on the door. In use, the universal roller bracket may secure the movable barrier to two tracks on either side of the movable barrier. If the tracks are angled with respect to a plane defined by the movable barrier in a closed position, the universal roller may account for the angle between the tracks and the movable barrier. It may account for this angle because each roller tube of the universal roller bracket may space a roller at a different distance away from the plane defining the movable barrier and/or from the movable barrier itself. Thus, the same universal roller bracket may be used at varying sections of the movable barrier to space rollers at gradually increasing distances from the movable barrier as height along the movable barrier increases. For example, at a lower section of the movable barrier, a roller securing the movable barrier to the track may be placed within a roller tube positioned closest to the movable barrier. At a location above this lower section, a roller may be placed within a roller tube spaced by a greater distance from the movable barrier. At another higher location, a roller may be placed in a roller tube spaced by a still greater distance from the movable barrier, and so on.

[0019] FIG. 1 is a perspective illustration of an example movable barrier system **100**, according to aspects of the present disclosure. In this example, the movable barrier is an upward acting garage door. In some examples, the movable barrier may be a sectional-type garage door. FIG. 1 illustrates a movable barrier **190** and a barrier operator **95**. In some implementations, the movable barrier **190** may include multiple sections **192**, multiple panels **195**, and multiple universal roller brackets **228**. In some implementations, the sections **192** make up the movable barrier **190**. For example, in FIG. 1, the movable barrier **190** may include four sections **192** arranged vertically. The panels **195** may

be positioned within the sections **192** and may be transparent or non-transparent.

[0020] In some implementations, the movable barrier system **100** described herein may be referred to as a barrier system, a door system, a garage door system, a gate system, or any other similar term. In some implementations, the movable barrier **190** may be referred to as a barrier, a door, a garage door, a sectional garage door, an upward acting garage door, a gate, a movable gate, a sliding gate, or any other similar term. In some implementations, the barrier operator **95** may alternatively be referred to as an operator, a door operator, a garage door operator, a gate operator, an opener, a door opener, a garage door opener, a gate opener, a control system, or any other similar term. In some implementations, the light fixture **118** may be referred to as a light, a light system, or any other similar term.

[0021] FIG. **1** shows that the movable barrier **190** provides access to a space or a room having a ceiling **117** and the light fixture **118** that is spaced from the barrier operator **95**. The movable barrier **190** may provide selective access to the space. The barrier operator **95** may be any suitable type of barrier operator. For example, in some implementations, the barrier operator **95** may be a jackshaft operator. In other implementations, the barrier operator **95** may be a direct drive wall or ceiling mounted operator, a belt driven operator, a chain driven operator, a screw drive operator, a trolley operator, a carriage operator, or any other type of barrier operator. The barrier operator **95** may include any suitable components. As shown in FIG. **1**, the barrier operator **95** may be disposed adjacent the movable barrier **190**. For example, in the implementation shown, the barrier operator **95** may be positioned on the same wall as the opening covered by the movable barrier **190**.

[0022] However, the barrier operator **95** may be positioned at any other location within the room shown in FIG. **1**. For example, the barrier operator **95** may be affixed to the ceiling **117**. In some implementations, the barrier operator **95** may be positioned on a different wall of the room or on the floor of the room. In some implementations, particularly in an implementation in which the barrier operator **95** is affixed or otherwise positioned on the ceiling **117** of the room, the light fixture **118** may be attached to, or a part of, the barrier operator **95**.

[0023] Any suitable structures or components may be implemented to facilitate movement of the movable barrier **190** between a closed position and an open position. In the example shown in FIG. **1**, the movable barrier **190** may be moved along one or more tracks **140**. Additionally shown in FIG. **1** is a shaft **130**, cable drums **132**, and a torsion spring **138**.

[0024] FIG. **1** illustrates the movable barrier **190** as an upward acting sectional door being movable between open and closed positions along the tracks **140**. The tracks **140** may be affixed to either side of the opening of the movable barrier **190**. In some implementations, the tracks **140** may be affixed to the wall of the room shown in FIG. **1** and/or the ceiling **117**. In some implementations, the movable barrier **190** may include one or more rolling or sliding components on either side sized and shaped to fit within and move in a longitudinal direction along the tracks **140**. The rolling or sliding components may be affixed to the brackets positioned on either side of the movable barrier **190**, such as the universal roller brackets **228** shown, and as will be described in more detail hereafter.

[0025] Components of the barrier system **100** shown in FIG. **1** may include any other suitable components. For example, the barrier system **100** may include rollers positioned on the movable barrier **190** or the tracks **140**. The system **100** may include sensors, such as safety sensors configured to detect the presence or motion of an object or person, seals positioned along any portion of the movable barrier **190** or the corresponding opening, tracks, cables, or tube shafts. The system may include extension springs to further reduce necessary rotational force of a motor, a motor rail, belts, motor head, motor arms, lift handles for manual operation, emergency release ropes, or any other suitable components.

[0026] It is noted that the movable barrier **190** may include any number of sections (e.g., sections **192a-d**), including a number of sections greater or less than those shown. Additionally, any of the sections **192a-d** may include any number of roller brackets of various types. As shown in FIG. **2**,

the movable barrier **190** may be a four-section barrier, such as the movable barrier **190** on a straight vertical track.

[0027] FIG. **2** is a side view of the movable barrier **190** in a closed position along the track **140** with universal roller brackets **228**, according to aspects of the present disclosure. As shown in FIG. **2**, while in a closed position, the movable barrier **190** may be positioned vertically. For example, FIG. **2** depicts two axes: a vertical axis **298** and a horizontal axis **299**. The movable barrier **190** may be positioned substantially parallel to the vertical axis **298**. Multiple universal roller brackets **228** may be positioned in relation to the movable barrier **190** securing the movable barrier **190** to the track **140** shown. As will be described in more detail below, the universal roller brackets **228** provide multiple roller tubes which each may receive a roller **230**. The roller tubes may be positioned at different offset lengths from the inner surface of the movable barrier **190**. Due to the multiple roller tubes placed at differing positions, the same universal roller bracket **228** may be used at multiple locations along the inner surface of the movable barrier **190** or may be affixed to different sections **192** of the movable barrier **190** and provide an angle **297** of the track **140**.

[0028] As shown in FIG. **2**, the track **140** of the movable barrier system may be positioned at an angle **297** from the vertical axis. In some implementations, this angled track **140** is oriented at varying distances from the jamb of the movable barrier **190**. Such an orientation of the track **140** ensures that the movable barrier **190** quickly pulls away from the jamb when the movable barrier **190** translates upward. It is noted that the angle **297** depicted in FIG. **2** may not be geometrically accurate, as shown. Rather, the depiction of the angle **297** may be exaggerated for pedagogical purposes (e.g., to show an increased offset distance between the movable barrier **190** and the track **140** with height).

[0029] It is noted that the movable barrier **190** may include any number of sections **192**, including a number of sections greater or less than those shown in FIG. **1** or FIG. **2**. Additionally, any of the sections **192** may include any number of roller brackets of various types. As shown in FIG. **2**, a four-section barrier, such as the movable barrier **190** may include five roller brackets including a lower roller bracket **221** at the bottom of the door and multiple universal roller brackets **228**. These roller brackets may be symmetrical and can be mounted on the left or right ends of the movable barrier.

[0030] In the implementation shown, the section **192a** of the movable barrier **190** is secured to the track **140** via a universal roller bracket **228a**. The sections **192b-192d** are similarly secured to the track **140** via respective universal roller brackets **228b-d**. As shown, the lower roller bracket **221** may affix a lower portion of the section **192a** to the track **140** as described previously.

[0031] FIG. **2** provides an enlarged view **290** of the universal roller bracket **228b**. As shown in the view **290**, the roller bracket **228** may include an upper bracket section including a protrusion **221b** and a protrusion **222b**. The protrusions **221b** and **222b** may be sized and shaped to be received within corresponding grooves **211b** and **212b** within an inner surface of the section **192b** shown. In this way, prior to installation and/or during installation of the universal roller bracket **228b**, the bracket **228b** may be moved along the grooves **211b** and **212b** in a direction horizontal to the movable barrier **190** (e.g., perpendicular to the horizontal axis **299** and the vertical axis **298**).

[0032] An upper arm **245** extends from the upper bracket section (see e.g., the upper section **346** of FIG. **3** and/or the upper section **446** of FIG. **4**) away from the inner surface of the section **192b** and supports a plurality of roller tubes. The roller tubes shown include a roller tube **223**, a roller tube **224**, a roller tube **225**, and a roller tube **226**. As shown, each of the roller tubes **223-226** may be spaced from the inner surface of the section **192b** or from the leg of the roller bracket **228** by a different offset length (e.g., lengths **L1**, **L2**, **L3**, and **L4**). For example, the roller tube **223** may be spaced from the inner surface of the movable barrier **192** and/or the upper bracket section of the roller bracket **228b** by an offset length **L1**. The roller tube **224** may be spaced from the inner surface of the movable barrier **192** and/or the upper bracket section of the roller bracket **228b** by an offset length **L2**. The roller tube **225** may be spaced from the inner surface of the movable barrier

192 and/or the upper bracket section of the roller bracket **228b** by an offset length **L3**. The roller tube **226** may be spaced from the inner surface of the movable barrier **192** and/or the upper bracket section of the roller bracket **228b** by an offset length **L4**. In some implementations, the offset lengths **L1-L4** between the roller tubes **223-226** and the section **192b** may provide the desired angle **297** between the movable barrier **190** and the track **140**.

[0033] Specifically, the roller tube **223** may be positioned at an offset length **L1** such that, with the roller bracket **228a** affixed to the lowest section **192a** of the movable barrier **192**, a corresponding roller **230** is positioned within the roller tube **223**. Such an arrangement is illustrated by the universal roller bracket **228a** shown in FIG. 2.

[0034] Similarly, the roller tube **224** may be positioned at an offset length **L2** such that, with the roller bracket **228b** affixed to the section **192b** of the movable barrier **192**, the roller **230** shown in the enlarged view **390** is positioned within the roller tube **224**. Such an arrangement is also illustrated by the universal roller bracket **228b** shown in FIG. 2.

[0035] The roller tube **225** may be positioned at an offset length **L3** such that, with the roller bracket **228c** affixed to the section **192c** of the movable barrier **192**, a corresponding roller **230** is positioned within the roller tube **225**. Such an arrangement is also illustrated by the bracket assembly **192c** shown in FIG. 2.

[0036] The roller tube **226** may be positioned at an offset length **L4** such that, with the roller bracket **228d** affixed to the section **192d** of the movable barrier **192**, a corresponding roller **230** is positioned within the roller tube **226**. Such an arrangement is also illustrated by the bracket assembly **228d** shown in FIG. 2.

[0037] It is noted that in some cases, the universal roller bracket **228d** shown and described with reference to FIG. 2 may alternatively be an upper roller bracket. For example, similar to the lower roller bracket **221**, an upper roller bracket may include a single roller tube spaced at a predetermined offset length (e.g., length **L4**) from the upper region of the section **192d**. In such a case, the universal roller brackets **228** may alternatively include only three roller tubes (e.g., roller tubes **223**, **224**, and **225**). Additional aspects of such a universal roller bracket are described in more detail with reference to FIG. 3 below.

[0038] The roller bracket **228b** additionally includes a lower arm **247** and a lower bracket section (see e.g., the lower section **344** of FIG. 3 and/or the upper section **444** of FIG. 4). Similar to the upper arm **245**, the lower arm **247** may support the roller tubes **223-226**. For example, the lower arm **237** may extend from the lower bracket section affixed to the section **192b** at an angle and is fixedly coupled to any one or more of the roller tubes **223-226**. In some implementations, the lower bracket section may include one or more openings through which a fastener, such as the fastener **227** shown, may be positioned. In some implementations, the fastener **227** may be a self-tapping fastener which creates and engages with a recess in the inner surface of the section **192b**. In other implementations, recesses may be formed within the section **192b** which may be sized and shaped to receive a corresponding fastener, such as the fastener **227**. It is anticipated that any suitable type of fastener may secure the lower bracket section to the inner surface of the section **192b**, including, for example, grooves preformed into the movable barrier section **192b** and corresponding protrusions of the universal bracket **228b**, similar to those described with reference to the upper bracket section.

[0039] As described above, the upper bracket section of the roller bracket **228** may be slidably coupled to the inner surface of the section **192b** while the lower bracket section may be fixedly coupled to the inner surface of the section **192b** by a fastener **227**. In some implementations, the upper portion may be slidably connected by the protrusions **221b** and **222b** and corresponding grooves **211b** and **212b** during installation at some time before the lower bracket section is fixedly secured to the section **192b** by the fastener **227**.

[0040] Because of the varying offset lengths between the roller tubes **223-226** of the roller bracket **228**, the same bracket **228** may be used to secure any of the sections **192a-d** of the movable barrier

192 by using different roller tubes for different sections as shown. This versatility of the roller bracket **228** reduces the number of unique parts required for manufacturing or installation of the barrier system shown and described with reference to FIG. 2. For example, the disclosed universal roller bracket **228** may reduce the number of unique roller brackets for a movable barrier from five (e.g., the lower roller bracket **221**, as well as four different sized roller brackets) to two (e.g., the lower roller bracket **221** and universal roller brackets **228**) or, in some cases three (e.g., the lower roller bracket **221**, universal roller bracket **228**, and an upper roller bracket) as described previously. In addition, by reducing the number of required unique components for a movable barrier, the universal roller bracket **228** reduces the number of unique parts required to be transported to an installation or servicing procedure. The reduction in unique components also reduces installation and service complexity. In addition, the universal roller bracket **228** may be symmetrical for use on the left and right ends of the section panels. The reduced number of unique parts reduces cost and complexity of manufacturing, increasing affordability and availability of the part. The design of the universal roller bracket **228** additionally reduces required manufacturing labor. In some implementations, the universal roller bracket **228** may be constructed of aluminum and may be extruded and cut to the desired length. In other implementations, the universal roller bracket **228** may be constructed of any other suitable material, such as steel. The part is made from an aluminum extrusion cut to desired length. As shown, the universal roller bracket **228** advantageously reduces manufacturing costs, including retooling costs, as well as inventory management and logistical cost and complexity.

[0041] As shown in the enlarged view **290** of FIG. 2, the position of the roller tubes **223-226** may be oriented along a line **295** extending away from the section **192b**. For example, the line **295** may extend from an upper region of the inner surface of the section **192b** of the movable barrier **192** downward and away from the movable barrier **192**. In some implementations, the line **295** may extend away from the movable barrier **192** at an angle **294**. In some implementations, the line **295** may be a straight line. In some implementations, such as the one shown in FIG. 2, the line **295** may be curved. For example, the roller tubes **223-226** may be positioned along the line **295** according to a curve of a curvature **296**. This curvature may arc back toward the inner surface of the section **192b**.

[0042] As mentioned previously, the positions of the roller tubes **223-226** may be arranged along a straight line or a curved line extending from the inner surface of the movable barrier **192**. Specifically, in an implementation in which the roller tubes **223-226** are arranged along a straight line, the locations of the roller tubes **223-226** may be described as having a linear relationship. For example, a linear relationship may relate to an increase in separation distance of the roller tubes being constant. As shown in FIG. 2, the difference between the offset length **L1** corresponding to the roller tube **223** and the offset length **L2** corresponding to the roller tube **224** may be the same as the difference in length between **L2** and **L3** as well as **L3** and **L4**.

[0043] For a linear or non-linear relationship between the roller tubes **223-226**, the difference between the lengths **L1-L4** may not be constant. For example, the difference in length between lengths **L2** and **L3** may be less than the difference in length between lengths **L1** and **L2**. Similarly, the difference in length between lengths **L3** and **L4** may be less than the difference in length between lengths **L2** and **L3**. In some implementations, a linear or non-linear relationship of the roller tubes **223-226** may account for vertical variation of the roller tubes **223-226** thus more precisely maintaining the angle **297** at all locations along the movable barrier **192**.

[0044] In some implementations, a hinge or joint may join the sections of the movable barrier **192**. For example, a hinge or joint may be positioned between the sections **192b** and **192c** and above the universal roller bracket **228b**. In some implementations, the orientation of the roller tubes **223-226** may provide a range of motion for the hinge or joint joining the sections **192b** and **192c** as the movable barrier **192** moves through curved sections of the track **140**.

[0045] The upper arm **245**, the lower arm **247**, and the roller tubes **223-226** may together form a

gap **250** between the inner surface of the section **192b** and a middle portion of the roller bracket **228b** between the upper and lower bracket sections. This gap **250** may advantageously reduce the amount of required material to construct the universal roller bracket **228b**. In addition, this gap **250** may provide additional flexibility to the universal roller bracket **228b**. Such increased flexibility may reduce the risk of breakage of the universal roller bracket **228b** as well as provide for smoother movement and/or operation of the movable barrier. It is also noted that a method of assembly of the universal roller bracket **228b** to the movable barrier **192** and a roller **230** may be substantially similar to the method of assembly of the universal roller bracket **328** described herein. [0046] FIG. 3 is a perspective view of a section **192** of the movable barrier **192**, a universal roller bracket **328**, and a roller **230**, according to aspects of the present disclosure. As shown in FIG. 3, the section **192** of the movable barrier **190** includes two grooves **211** and **212**, like the grooves **211b** and **212b** described previously. The grooves **211** and **212** may extend in a horizontal direction along the section **192** of the movable barrier **190**.

[0047] The universal roller bracket **328** may be similar to the universal roller bracket **228** described with reference to FIG. 2. For example, the universal roller bracket **328** may include an upper bracket section **346**. The upper bracket section **346** may include two protrusions: a protrusion **321** and a protrusion **322**. The protrusions **321** and **322** may be positioned within the corresponding grooves **211** and **212** of the section **192** of the movable barrier **190**. In this way, the universal roller bracket **328**, with the protrusions **321** and **322** within the grooves **211** and **212** may be permitted to move in a horizontal direction **390** along the section **192** of the movable barrier **190** but may be constrained from moving in any other direction.

[0048] The universal roller bracket **328** may include three roller tubes including a roller tube **323**, a roller tube **324**, and a roller tube **325**. Similar to the roller tubes **223-226** described with reference to FIG. 2, each of the roller tubes **323-325** may be sized and shaped to receive a roller **230**. For example, the roller **230** shown may include a circular portion **232** of a larger diameter, and a roller stem **234**. The roller stem **234** may be of cylindrical shape with a smaller diameter than the portion **232**. The roller stem **234** may be inserted within any of the roller tubes **323-325** and/or any of those previously or hereafter described.

[0049] The universal roller bracket **328** may additionally include a lower bracket section **344**. The lower bracket section **344** may include two openings **340** and **342**. In some implementations, the openings **340** and **342** may receive fasteners used to secure the lower bracket section **344** to the section **192** of the movable barrier **190**. In some implementations, the recesses **340** and **342** may align with corresponding orifices within the section **192** of the movable barrier **192** also sized and shaped to receive fasteners.

[0050] During an assembly of the universal roller bracket **328** (as well as any other brackets described herein) with the movable barrier **190** and the roller **230**, the circular portion **232** of the roller **230** may be inserted within a track of a movable barrier system (e.g., the track **140** described with reference to FIGS. 1-3). In some assemblies, a roller **230** may be inserted within a left track and an additional roller **230** may be inserted within a corresponding right track.

[0051] At an additional step of assembly, the universal roller bracket **328** may be affixed to the section **192** of the movable barrier **190** by sliding the protrusions **321** and **322** of the upper bracket section **346** into the corresponding grooves **211** and **212** of the movable barrier **190**. Similarly, a left universal roller bracket may be positioned at a left side of the movable barrier **190** and a right universal roller bracket may be positioned at a right side of the movable barrier **190**. In some implementations, the protrusions **321** and **322** (or any other protrusions described herein) may alternatively be inserted into corresponding grooves **211** and **212**.

[0052] At an additional step of assembly, the movable barrier **190** may be positioned relative to the left and right tracks such that the rollers **230** in either track align with one of the roller tubes (e.g., the roller tubes **323-325**) of the left and right universal roller brackets.

[0053] At an additional step of assembly, the left and right universal roller brackets may be slid

outward toward an outer left and right edge of the movable barrier **190** such that the appropriate roller tube of the brackets is positioned around the stems of the rollers **230**. In some implementations, at any time after the roller **230** engages the track (e.g., the track **140** of FIGS. **1-2**), the roller bracket may slide toward the end of the section panel engaging the roller stem **234** as it slides into the appropriate roller tube. As previously described, which roller tube of the bracket receives the stem of the rollers **230** may depend on the height of the universal roller brackets and/or which section of the movable barrier (e.g., the sections **192a-d** of the movable barrier **190** of FIG. **2**) the brackets are affixed to.

[0054] After the roller tubes are inserted into both the left and right tracks as well as the universal roller brackets, one or more fasteners may be placed through the recesses of the lower bracket sections of the universal roller brackets securing the universal roller brackets in place and preventing movement of the rollers **230** out of the corresponding roller tubes.

[0055] FIG. **4** is a perspective view of a section **192** of the movable barrier **190**, a universal roller bracket **428**, and a roller **230**, according to aspects of the present disclosure. The universal roller bracket **428** may be similar to the universal roller bracket **228** described with reference to FIG. **2** and/or the universal roller bracket **328** described with reference to FIG. **3**. For example, the universal roller bracket **428** may include an upper bracket section **446**. The upper bracket section **446** may include two protrusions: a protrusion **421** and a protrusion **422**. The protrusions **421** and **422** may be positioned within the corresponding grooves **211** and **212** of the movable barrier **190**. In this way, the universal roller bracket **428**, with the protrusions **421** and **422** within the grooves **211** and **212** may be permitted to move in the horizontal direction **390** along the panel of the movable barrier **190** but may be constrained from moving in any other direction.

[0056] Like the universal roller bracket **328**, the universal roller bracket **428** may include three roller tubes including a roller tube **423**, a roller tube **424**, and a roller tube **425**. Similar to the roller tubes **223-226** described with reference to FIG. **2** and/or the roller tubes **323-325** with reference to FIG. **3**, each of the roller tubes **423-425** may be sized and shaped to receive a roller **230**.

[0057] The universal roller bracket **428** may additionally include a lower bracket section **444**. The lower bracket section **444** may include two openings **440** and **442**. In some implementations, the recesses **440** and **442** (not pictured) may receive fasteners used to secure the lower bracket section **444** to the section **192** of the movable barrier **190**. In some implementations, the openings **440** and **442** may align with corresponding orifices within the section **192** of the movable barrier **190** also sized and shaped to receive fasteners.

[0058] A method of assembly of the universal roller bracket **428** may be similar to a method of assembly of the universal roller bracket **328**.

[0059] The roller tubes **423-425** of the universal roller bracket **428** may be spaced from an inner surface of the movable barrier **192** by an upper arm **445** and a lower arm **447**. As shown in FIG. **4**, the roller tubes **423-425** may be spaced from the movable barrier **190** by an offset length that is greater than the offset length of the roller tubes **323-325** described with reference to FIG. **3**. As a result, a gap between the roller tubes **423-425** and the movable barrier **190** may be larger than a gap formed between the roller tubes **323-325** and the movable barrier **190**.

[0060] In some implementations, the arm **445** and the roller tubes **423-425** may be positioned along a line extending from the movable barrier **190** similar to the line **295** described with reference to FIG. **2**. For example, the arm **445** and roller tubes **423-425** may be spaced from an inner surface of the movable barrier **190** at the angle **297** and may be positioned along a straight line or a curved line of curvature **296** (FIG. **2**).

[0061] Because the universal roller bracket **428** spaces the roller tubes **423-425** away from the inner surface of the movable barrier **190** by a greater distance than the other universal roller brackets described previously, the universal roller bracket **428** may be used for securing taller movable barriers with additional sections to the movable barriers described previously.

[0062] In one example, a movable barrier may include six sections similar to the sections **192a-d**

described with reference to FIG. 2. The lower roller bracket **221** may secure a lower region of the lowest section (e.g., the section **192a**) to a corresponding track (e.g., a track similar to the track **140**). The universal roller bracket **328** may be used to secure the lowest three sections of the movable barrier, with rollers **230** inserted into the three different roller tubes **323-325** of the universal roller bracket **328** according to the position of the respective barrier section. Three additional sections may be secured to the corresponding track by the universal roller bracket **428** using the roller tubes **423-425** according to the position of the respective barrier section. In this way, the angle **297** (FIG. 2) between the movable barrier and the track may be maintained even for taller movable barriers.

[0063] Persons of ordinary skill in the art will appreciate that the implementations encompassed by the present disclosure are not limited to the particular exemplary implementations described above. In that regard, although illustrative implementations have been shown and described, a wide range of modification, change, combination, and substitution is contemplated in the foregoing disclosure. It is understood that such variations may be made to the foregoing without departing from the scope of the present disclosure. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the present disclosure.

Claims

1. A universal roller bracket for a movable barrier having a surface, the universal roller bracket comprising: a first bracket section configured to be coupled to the movable barrier; a first arm extending from the first bracket section; and a plurality of roller tubes supported by the first arm and spaced from the surface of the movable barrier by the first arm, each roller tube of the plurality of roller tubes configured to receive a roller stem, each roller tube being spaced a different distance from the surface of the movable barrier.
2. The universal roller bracket of claim 1, wherein the first bracket section is configured to be slidably coupled to the surface of the movable barrier.
3. The universal roller bracket of claim 2, wherein the first bracket section includes a protrusion configured to be positioned within a complementary groove of the surface of the movable barrier.
4. The universal roller bracket of claim 1, further comprising a second bracket section spaced from the first bracket section, the second bracket section configured to be fixedly coupled to the surface of the movable barrier with a fastener.
5. The universal roller bracket of claim 1, wherein the distance of the roller stem from the surface of the movable barrier is based on an installation height of the roller stem when the movable barrier is in a closed position.
6. The universal roller bracket of claim 1, wherein the different distances of each roller tube correspond to a non-linear relationship.
7. The universal roller bracket of claim 1, further comprising a second arm, the first arm and the second arm forming a gap between the plurality of roller tubes and the surface of the movable barrier.
8. The universal roller bracket of claim 2, wherein the first bracket section includes a protrusion configured to be inserted into a complementary groove of the surface of the movable barrier.
9. A movable barrier system, comprising: a first section of a movable barrier comprising an inner surface and an outer surface, the inner surface comprising a retaining groove; a first universal roller bracket comprising: a first bracket section comprising a protrusion configured to be positioned within the retaining groove of the first section; and a first plurality of roller tubes spaced from the inner surface, wherein each roller tube is positioned a different distance from the inner surface; and a roller comprising a roller stem, wherein the roller stem is configured to be positioned within one of the plurality of roller tubes.
10. The movable barrier system of claim 9, further comprising a track angled relative to a vertical

axis of the first section.

11. The movable barrier system of claim 10, wherein the movable barrier is configured to be substantially vertical when in a closed position when the roller stem is positioned within the one roller tube of the plurality of roller tubes.

12. The movable barrier system of claim 9, wherein the first universal roller bracket includes a second bracket section configured to be fixedly coupled to the first section.

13. The movable barrier system of claim 9, further comprising: a second section of the movable barrier comprising an inner surface and an outer surface; and a second universal roller bracket including a second plurality of roller tubes, the second universal roller bracket configured to be coupled to the second section, wherein each roller tube of the second plurality of roller tubes is positioned a different distance from the inner surface of the second section, and wherein the distance of each roller tube of the second plurality of roller tubes from the inner surface of the second section is different than the distance of each roller tube of the first plurality of roller tubes of the first universal roller bracket from the inner surface of the first section.

14. The movable barrier system of claim 13, wherein the second section is positioned above the first section and the second universal roller bracket is positioned above the first universal roller bracket when the movable barrier is in a closed position.

15. A method of assembling a movable barrier system, the method comprising: positioning a roller within a track of the movable barrier system; slidably coupling a first surface of a universal roller bracket to a section of a movable barrier, the universal roller bracket comprising a plurality of roller tubes spaced from the section of the movable barrier, wherein each roller tube of the plurality of roller tubes is spaced from the section by a different distance; sliding the universal roller bracket to a position near an end of the section of the movable barrier; inserting a roller stem of the roller within one roller tube of the plurality of roller tubes; and fixedly coupling a second surface of the universal roller bracket to the section of the movable barrier.

16. The method of claim 15, further comprising fixedly coupling the first surface of the universal roller bracket to the section of the movable barrier.

17. The method of claim 15, wherein slidably coupling the first surface of the universal roller bracket to the section of the movable barrier includes positioning a protrusion of the universal roller bracket within a corresponding groove of the section of the movable barrier.

18. The method of claim 15, wherein coupling the second surface of the universal roller bracket to the section of the movable barrier includes coupling the second surface of the universal roller bracket to the section of the movable barrier with a fastener.

19. The method of claim 15, wherein the one roller tube of the plurality of roller tubes into which the roller stem is inserted is selected based on an installation height of the universal roller bracket on the section of the movable barrier.

20. The method of claim 19, wherein the installation height of the universal roller bracket is determined when the movable barrier is in a closed position.
